

Nonlinear Regression Functions

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Chapter Outline

- General Strategy for Nonlinear Regression Functions.
- Polynomial regression functions of a single Regressor.
- Logarithmic regression functions of a single Regressor.
- Interactions between independent variables.

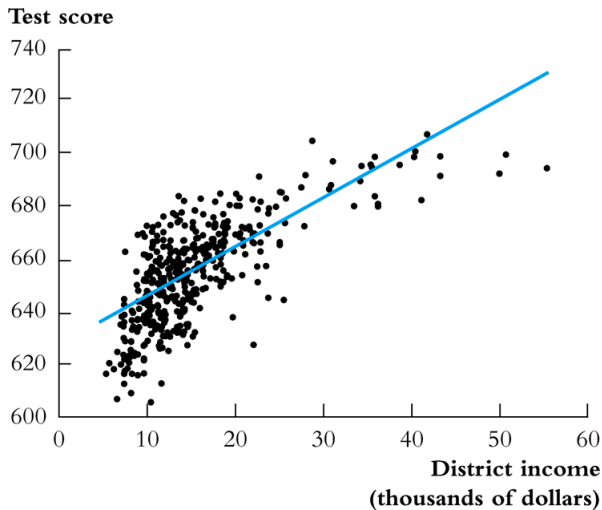
Motivation

- Until now we have assumed that the population regression function was linear.
- In a linear regression the slope coefficients are constant (i.e., do not depend on the value of X).
- We are going to analyze two departures from the above assumption
 - ① The effect on Y of a change in X_j depends on the the value of X_j .
 - ② The effect on Y of a change in X_j depends on the value of another regressor X_h .
- Our population regression model will be nonlinear in the regressors, this is $E[Y_i|X_{1i}, \dots, X_{Ki}]$ is a nonlinear function of the X 's.
- However, our population regression function will be linear on the unknown population coefficients and therefore we can estimate them using *OLS*.

Case Study

- We have found that the economic background of the students is an important factor in explaining performance on standardized tests.
- So far we used two economic background variables: the percentage of families qualifying for subsidized lunch, and the percentage of families qualifying for income assistance.
- Here, we are going to replace these two variables for a more aggregate variable: average annual income per-capita in the district.
- Inspection of the income data suggest a nonlinear relationship between test scores and income implying that a straight line is not a good representation of the relationship.

Case Study



Case Study

Consider instead the following quadratic population regression

$$TestScore_i = \beta_0 + \beta_1 Income + \beta_2 Income_i^2 + u_i \quad (1)$$

The equation above is called the *quadratic regression model* because the population regression model

$$E[TestScore_i | Income] = \beta_0 + \beta_1 Income_i + \beta_2 Income_i^2$$

is a quadratic function of the independent variable *Income*.

Because the quadratic regression model is a variant of the multiple regression model already discussed, we can estimate it using *OLS*.

Case Study

Estimating our quadratic model using *OLS* gives

$$\widehat{TestScore} = 607.3 + 3.85 \text{ Income} - 0.0423 \text{ Income}^2, \quad \bar{R}^2 = 0.554 \quad (2)$$

(2.9) (0.27) (0.0048)

Note that we can test if the linear specification is true against the alternative that the quadratic specification is true by testing $H_0 : \beta_2 = 0$ against $\beta_2 \neq 0$.

A General Strategy for Modeling Nonlinear Regression Functions

If a relation between Y and X is nonlinear:

- The effect on Y of a change in X depends on the value of X –that is, the marginal effect of X is not constant.
- A linear regression is mis-specified: the functional form is wrong.
- The estimator of the effect on Y of X is biased: in general it is not even right on average.
- The solution is to estimate a regression function that is nonlinear in X .

A General Strategy for Modeling Nonlinear Regression Functions

General Nonlinear Population Model

The nonlinear population regression model is

$$Y_i = f(X_{1i}, \dots, X_{Ki}) + u_i, \text{ for } i = 1, \dots, n \quad (3)$$

where $f(X_{1i}, \dots, X_{Ki})$ is the population nonlinear regression function.

A General Strategy for Modeling Nonlinear Regression Functions

Assumptions

- 1 $E[u_i|X_{1i}, \dots, X_{Ki}] = 0$ (same as before); implies that f is the conditional expectation of Y given the X 's.
- 2 $(X_{1i}, \dots, X_{Ki}, Y_i)$ are i.i.d. (same).
- 3 Big outliers are rare (same idea; the precise mathematical condition depends on the specific f).
- 4 No perfect multicollinearity (same idea; the precise statement depends on the specific f).

A General Strategy for Modeling Nonlinear Regression Functions

Effect on Y of a Change in X

The effect on Y on a change in X_1 , ΔX_1 , holding $X_2, \dots, X_K$ constant is,

$$\Delta Y = f(X_1 + \Delta X_1, X_2, \dots, X_K) - f(X_1, X_2, \dots, X_K) \quad (4)$$

Since the regression function f is unknown, the population effect on Y of a change in X_1 is also unknown and must be estimated. The predicted change in Y is

$$\Delta \hat{Y} = \hat{f}(X_1 + \Delta X_1, X_2, \dots, X_K) - \hat{f}(X_1, X_2, \dots, X_K) \quad (5)$$

In the linear regression model coefficients β_j have a natural interpretation. In nonlinear models coefficients are best interpreted by calculating the predicted effect on Y of changing one or more of the independent variables.

A General Strategy for Modeling Nonlinear Regression Functions

Standard Errors of the Estimated Effects: Linear Regression

- Suppose we want to compute a confidence interval for the effect on Y , ΔY .
- It is easy to compute the standard error for $\Delta \hat{Y}$ when the regression is linear.
- The estimated effect of a change in X_1 is $\hat{\beta}_1 \Delta X_1$, so the 95% confidence interval for the change is $\left[\hat{\beta}_1 \Delta X_1 \pm 1.96 \times SE \left(\hat{\beta}_1 \right) \Delta X_1 \right]$.
- In the nonlinear model, computing the standard error of ΔY requires to use a technique introduced in chapter 7 for testing single restrictions involving multiple coefficients.

A General Strategy for Modeling Nonlinear Regression Functions

Standard Errors of the Estimated Effects: Nonlinear Regression

- To illustrate the procedure, consider the estimated change in test scores associated with a change in income from 10 to 11,

$$\Delta \hat{Y} = \hat{\beta}_1 (11 - 10) + \hat{\beta}_2 (11^2 - 10^2) = \hat{\beta}_1 + 21\hat{\beta}_2.$$

- The standard error of the predicted change is

$$SE(\Delta \hat{Y}) = SE(\hat{\beta}_1 + 21\hat{\beta}_2)$$

- Hence, to compute the standard error of $\Delta \hat{Y}$ we need to compute the standard error of $\hat{\beta}_1 + 21\hat{\beta}_2$.

A General Strategy for Modeling Nonlinear Regression Functions

Standard Errors of the Estimated Effects: Nonlinear Regression

To compute standard error of $\hat{\beta}_1 + 21\hat{\beta}_2$ we can use two approaches

- 1 Compute the F – *statistic* testing the null hypothesis that $\beta_1 + 21\beta_2 = 0$ and use

$$SE(\Delta\hat{Y}) = \frac{|\Delta\hat{Y}|}{\sqrt{F}} \quad (6)$$

- 2 We can transform the regressors so that in the transformed regression, one of the coefficients is $\beta_1 + 21\beta_2$.

A General Strategy for Modeling Nonlinear Regression Functions

A Guide

The general approach to modeling nonlinear regression functions has five elements

- 1 Identify a possible nonlinear relationship.
- 2 Specify a nonlinear function and estimate its parameters by *OLS*.
- 3 Determine whether the nonlinear model improves upon the linear model.
- 4 Plot the estimated nonlinear regression function.
- 5 Estimate the effect on Y of a change in X .

Nonlinear Functions of a Single Independent Variable

We will look at two complementary approaches

① *Polynomials in X*

The population regression function is approximated by a quadratic, cubic, or higher-degree polynomial.

② *Logarithmic transformations*

Y and/or X is transformed by taking its logarithm. This gives a “percentages” interpretation that makes sense in many applications.

Nonlinear Functions of a Single Independent Variable

Polynomials

Approximate the population regression function by a polynomial:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \dots + \beta_r X_i^r + u_i, \quad (7)$$

where r denotes the highest power of X . When $r = 2$ we have the quadratic regression model.

- This is just the linear multiple regression model –except that the regressors are powers of X !
- Estimation, hypothesis testing, etc. proceeds as in the multiple regression model using *OLS*.
- The coefficients are difficult to interpret, but the regression function itself is interpretable.

Nonlinear Functions of a Single Independent Variable

Which degree of polynomial should you use?

- Increasing the degree of the polynomial (i.e., increasing r) introduces more flexibility into the regression function and allows it to match more shapes; a polynomial of degree r can have up to $r - 1$ bends (that is, inflection points) in its graph.
- Increasing r means adding more regressors which can reduce the precision of the estimated coefficients.
- Thus, the answer balances a trade-off between flexibility and statistical precision.
- A practical way to determine the degree of the polynomial is to ask whether the coefficients in the polynomial associated to the largest values are zero.

Nonlinear Functions of a Single Independent Variable

Which degree of polynomial should you use?

- 1 Pick a maximum value of r and estimate the regression for that r .
- 2 Use the t -statistic to test individual statistical significance of β_r . If you reject, use the polynomial of degree r .
- 3 If you do not reject $H_0 : \beta_r = 0$, eliminate X^r , estimate the $r - 1$ degree polynomial and repeat *step 2*.
- 4 If you do not reject $H_0 : \beta_{r-1} = 0$, eliminate X^{r-1} , estimate the $r - 2$ degree polynomial and repeat the procedure until no more eliminations take place.
- 5 Note that the initial r must be chosen using (economic) theory. In most econ applications the initial r is set between 2 and 4.

Case Study

Lets look at the quadratic and cubic specifications

$$TestScore = \beta_0 + \beta_1 Income_i + \beta_2 Income_i^2 + u_i, \quad (8)$$

$$TestScore = \beta_0 + \beta_1 Income_i + \beta_2 Income_i^2 + \beta_3 Income_i^3 + u_i. \quad (9)$$

Case Study

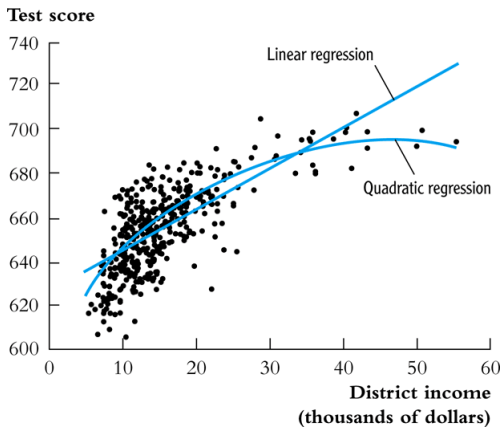
```
generate avginc2 = avginc*avginc;      Create a new regressor
reg testscr avginc avginc2, r;
```

Regression with robust standard errors

Number of obs = 420
F(2, 417) = 428.52
Prob > F = 0.0000
R-squared = 0.5562
Root MSE = 12.724

		Robust					
testscr		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----							
avginc		3.850995	.2680941	14.36	0.000	3.32401	4.377979
avginc2		-.0423085	.0047803	-8.85	0.000	-.051705	-.0329119
_cons		607.3017	2.901754	209.29	0.000	601.5978	613.0056

Case Study



Case Study

To interpret the estimated regression function consider

$$\widehat{TestScore} = \underset{(2.9)}{607.3} + \underset{(0.27)}{3.85} Income - \underset{(0.0048)}{0.0423} Income^2.$$

The predicted change in $\widehat{TestScore}$ for a change in income per-capita from \$5000 to \$6000 is

$$\begin{aligned}\Delta \widehat{TestScore} &= \\ &= 607.3 + 3.85 \times 6 - 0.0423 \times 6^2 \\ &\quad - (607.3 + 3.85 \times 5 - 0.0423 \times 5^2) \\ &= 3.4\end{aligned}$$

Case Study

Change in Income	$\Delta TestScore$
from 5 to 6	3.4
from 25 to 26	1.7
from 45 to 46	0.0

- The “effect” of a change in income is greater at low than high income levels (perhaps, a declining marginal benefit of an increase in school budgets?)
- Caution! What is the effect of a change from 65 to 66?
- Do not extrapolate outside the range of the data!

Case Study

```
gen avginc3 = avginc*avginc2;  
reg testscr avginc avginc2 avginc3, r;
```

Create the cubic regressor

Regression with robust standard errors

Number of obs = 420
F(3, 416) = 270.18
Prob > F = 0.0000
R-squared = 0.5584
Root MSE = 12.707

		Robust					
testscr		Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
-----+-----							
avginc		5.018677	.7073505	7.10	0.000	3.628251	6.409104
avginc2		-.0958052	.0289537	-3.31	0.001	-.1527191	-.0388913
avginc3		.0006855	.0003471	1.98	0.049	3.27e-06	.0013677
_cons		600.079	5.102062	117.61	0.000	590.0499	610.108

Nonlinear Functions of a Single Independent Variable

- Another way to specify a nonlinear regression function is to use the natural logarithm of Y or X .
- Logarithms convert changes in variables into percentage changes. This result relies on the fact that when Δx is small, the difference between the logarithm of $x + \Delta x$ and the logarithm of x is approximately $\Delta x/x$. This is,

$$\ln(x + \Delta x) - \ln(x) \approx \frac{\Delta x}{x},$$

when $\frac{\Delta x}{x}$ is small.

- Many relationships are naturally expressed in terms of percentages.

Nonlinear Functions of a Single Independent Variable

There are three different log regression specifications

	Case	Population Regression Model
I.	linear-log	$Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i$
II.	log-linear	$\ln(Y_i) = \beta_0 + \beta_1 X_i + u_i$
III.	log-log	$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i$

- The interpretation of the slope coefficient differs in each case.
- The interpretation is found by applying the general “before and after” rule: “figure out the change in Y for a given change in X .”
- Each case has a natural interpretation (for small changes in X).

Nonlinear Functions of a Single Independent Variable

Linear-log model

$$Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i \quad (10)$$

- Compute Y “before” and “after” changing X :

$$Y = \beta_0 + \beta_1 \ln(X) \text{ (before)}$$

$$Y + \Delta Y = \beta_0 + \beta_1 \ln(X + \Delta X) \text{ (after)}$$

- Subtract (“after”)- (“before”)

$$\Delta Y = \beta_1 [\ln(X + \Delta X) - \ln(X)] .$$

- Then, using the fact about logarithms

$$\Delta Y \approx \beta_1 \frac{\Delta X}{X} \text{ then } \beta_1 \approx \frac{\Delta Y}{\Delta X/X} .$$

Nonlinear Functions of a Single Independent Variable

Linear-log model

- Then, we have

$$Y_i = \beta_0 + \beta_1 \ln(X_i) + u_i,$$

- and

$$\beta_1 \approx \frac{\Delta Y}{\Delta X/X}.$$

- Note that $100 \times \frac{\Delta X}{X}$ is the percentage change in X , so a 1% increase in X (this is multiplying X by 1.01) is associated with a $0.01\beta_1$ change in Y .
- Thus 1% increase in $X \rightarrow 0.01$ increase in $\ln(X) \rightarrow 0.01\beta_1$ increase in Y .

Case Study

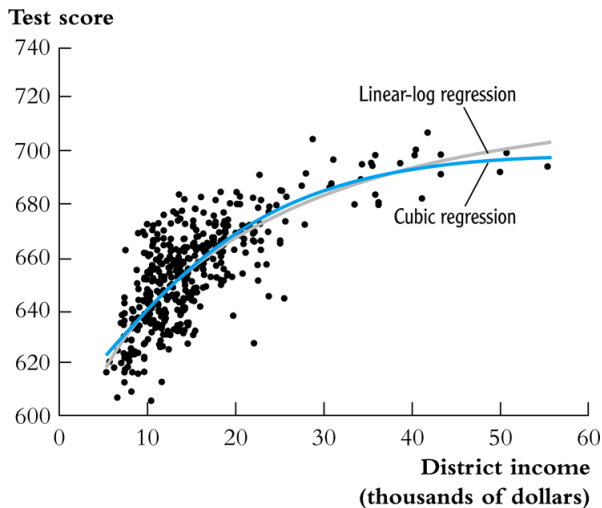
- First define the new regressor, $\ln(\text{Income})$.
- The model is now linear in $\ln(\text{Income})$, so the linear-log model can be estimated by *OLS*

$$\widehat{\text{TestScore}} = 557.8 + 36.42 \ln(\text{Income}), \quad \bar{R}^2 = 0.561. \quad (11)$$

(3.8) (1.4)

- Thus, a 1% increase in Income is associated with an increase in TestScore of 0.36 points on the test.
- Standard errors, confidence intervals, $\bar{R}^2$ –all the usual tools of regression apply here.

Case Study



Nonlinear Functions of a Single Independent Variable

Log-linear model

$$\ln(Y_i) = \beta_0 + \beta_1 X_i + u_i \quad (12)$$

- Compute Y “before” and “after” changing X :

$$\ln(Y) = \beta_0 + \beta_1 X \text{ (before)}$$

$$\ln(Y + \Delta Y) = \beta_0 + \beta_1(X + \Delta X) \text{ (after)}$$

- Subtract (“after”)- (“before”)

$$\ln(Y + \Delta Y) - \ln(Y) = \beta_1 \Delta X.$$

- Then, using the fact about logarithms

$$\frac{\Delta Y}{Y} \approx \beta_1 \Delta X \text{ then } \beta_1 \approx \frac{\Delta Y/Y}{\Delta X}.$$

Nonlinear Functions of a Single Independent Variable

Log-linear model

- Then, we have

$$\ln(Y_i) = \beta_0 + \beta_1 X_i + u_i.$$

- and

$$\beta_1 \approx \frac{\Delta Y / Y}{\Delta X}.$$

- Now $100 \times \frac{\Delta Y}{Y}$ is the percentage change in Y , so an increase in X by one unit (this is $\Delta X = 1$) is associated with a $100\beta_1\%$ change in Y .

Case Study

- First define the new regressor, $\ln(\text{TestScore})$.
- The model is linear in Income , so the log-linear model can be estimated by *OLS*

$$\ln(\widehat{\text{TestScore}}) = \underset{(0.003)}{6.439} + \underset{(0.00018)}{0.00284}\text{Income}, \quad \bar{R}^2 = 0.497. \quad (13)$$

- Thus, for every additional unit of income (in thousands of dollars), test scores are predicted to increase by 0.284%, (i.e., $[(100 \times 0.00284)\%]$).

Nonlinear Functions of a Single Independent Variable

Log-log model

$$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i \quad (14)$$

- Compute Y “before” and “after” changing X :

$$\ln(Y) = \beta_0 + \beta_1 \ln(X) \text{ (before)}$$

$$\ln(Y + \Delta Y) = \beta_0 + \beta_1 \ln(X + \Delta X) \text{ (after)}$$

- Subtract (“after”)- (“before”)

$$\ln(Y + \Delta Y) - \ln(Y) = \beta_1 [\ln(X + \Delta X) - \ln(X)] .$$

- Then, using the fact about logarithms

$$\frac{\Delta Y}{Y} \approx \beta_1 \frac{\Delta X}{X} \text{ then } \beta_1 \approx \frac{\Delta Y/Y}{\Delta X/X} .$$

Nonlinear Functions of a Single Independent Variable

Log-log model

- Then, we have

$$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i.$$

- and

$$\beta_1 \approx \frac{\Delta Y / Y}{\Delta X / X}.$$

- Now $100 \times \frac{\Delta Y}{Y}$ is the percentage change in Y , and $100 \times \frac{\Delta X}{X}$ is the percentage change in X .
- Then, a 1% change in X is associated with a $\beta_1\%$ change in Y .
- In the *log-log* specification, β_1 has the interpretation of an elasticity.

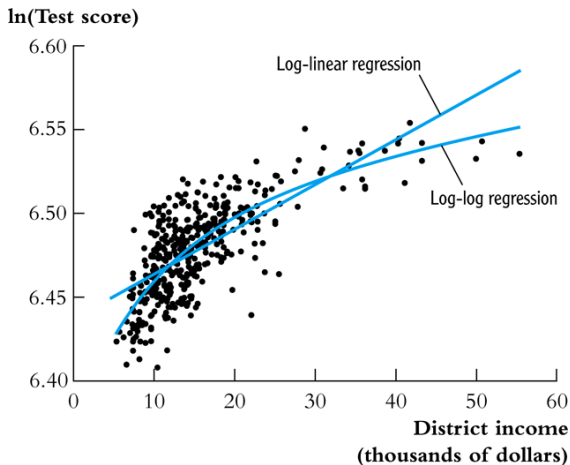
Case Study

- First define the new regressors, $\ln(\text{TestScore})$ and $\ln(\text{Income})$.
- The model of $\ln(\text{TestScore})$ in $\ln(\text{Income})$ is linear so the log-log model can be estimated by *OLS*

$$\ln(\widehat{\text{TestScore}}) = \underset{(0.006)}{6.336} + \underset{(0.0021)}{0.0554} \ln(\text{Income}), \quad \bar{R}^2 = 0.027. \quad (15)$$

- An 1% increase in *Income* is associated with an increase of .0554% in *TestScore*.
- Alternatively, the elasticity of the test score with respect to the price is 0.054.

Case Study



Case Study

- Visual inspection of the log-linear and log-log regressions suggest that neither seems to fit as well as the cubic or linear-log.
- Formal comparison is difficult because the dependent variables differ.
- To choose among alternative specifications we must rely on theory.

Nonlinear Functions of a Single Independent Variable

Summary

- Three cases, differing in whether Y and/or X is transformed by taking logarithms.
- The regression is linear in the new variable(s) $\ln(Y)$ and/or $\ln(X)$, and the coefficients can be estimated by *OLS*.
- Hypothesis tests and confidence intervals are now implemented and interpreted “as usual.”
- The interpretation of β_1 differs from case to case.
- The choice of specification (functional form) should be guided by judgement (which interpretation makes the most sense in your application?), tests, and plotting predicted values

Nonlinear Functions of a Single Independent Variable

Comparing the cubic and linear-log specifications

- The two estimated models are quite similar.
- The $\bar{R}^2$ of the logarithmic regression is 0.561 while that of the cubic regression is 0.555.
- The logarithmic specification is better behaved near the extremes of the sample, especially for large values of income.

Case Study

